

Full Length Article

Psychological drivers of compliance intention–The roles of professionals' motivational profiles and personality traits in following information systems security policies at work

Thomas Keller^{a,*}, Julia Warwas^a, Josef Guggemos^b, Verena Zengerle^a

^a University of Hohenheim, Department of Business Education, Fruwirthstr. 47, Stuttgart 70599, Germany

^b University of Education Schwäbisch Gmünd, Department of Vocational Education and Training, Oberbettringer Str. 200, Schwäbisch Gmünd 73525, Germany

ARTICLE INFO

Dataset link: [Motivational Qualities \(Original data\)](#)

Keywords:

Cybersecurity
Compliance
Personality traits
Motivation
Self-determination theory
Internalization

ABSTRACT

Cybersecurity is becoming increasingly important in organizations. Many data breaches and successful cyber-attacks occur when employees ignore, bypass, or inconsistently apply security rules. While existing research has focused on personality traits to explain compliance with information systems security policies (ISSP), it has largely neglected the domain-specific motivational regulation of behavior. Yet, sustaining effortful and sometimes inconvenient security practices depends not only on stable personality traits but also on how employees motivationally regulate their behaviors in this domain, highlighting the need to consider both constructs when explaining ISSP compliance intention. The present study addresses this gap by using the five-factor model of personality and self-determination theory (SDT) to examine how stable traits and distinct motivational qualities relate to ISSP compliance intention. In a cross-sectional online survey, 218 office employees from small and medium-sized enterprises reported their personality traits, security-related work motivation, and ISSP compliance intention. We tested the multidimensional structure of the adapted motivation measure using confirmatory factor analysis and identified qualitatively distinct motivational profiles through latent profile analysis. To examine the incremental contributions of structural characteristics, personality traits, and motivational profiles to ISSP compliance intention, we conducted hierarchical multiple regression analyses. The results support a multidimensional structure of motivational regulation and yield three profiles that differ in their extent of self-regulated behavior in pursuit of internalized goals. Structural characteristics (company size) and personality traits (agreeableness, conscientiousness) were significantly associated with ISSP compliance intention ($R^2 = 0.257$). Motivational profiles accounted for additional variance ($\Delta R^2 = 0.08$; final $R^2 = 0.333$), with autonomous profiles (i.e., profiles characterized by internalized and self-endorsed reasons for secure behavior) reporting higher ISSP compliance intention than the amotivated profile. Based on these findings, the study outlines how SDT-based interventions can support employees' psychological needs for autonomy, competence, and social relatedness, thereby fostering more internalized and enduring ISSP compliance.

1. Introduction

The digital transformation has led to a notable increase in cyber-crime on an annual basis (ENISA, 2021). Concurrently, the sophistication of attacks is evolving (Adriko and Nurse, 2026; Warkentin and Willison, 2009), with a particular focus on targeting employees, who are perceived as the most vulnerable link in the cybersecurity chain (Gallo et al., 2024; Rahman et al., 2025). By employing social engineering techniques such as phishing (Butavicius et al., 2022), cybercriminals exploit the trust, carelessness, or ignorance of employees to gain access

to sensitive information or systems (Humayun et al., 2020). Those who fall victim to such attacks are frequently unaware of the potential dangers, yet the consequences can be extensive. In addition to the financial damage (IBM Security, 2024; Jouini et al., 2014), a successful cyber-attack can also result in considerable reputational damage, as the trust of customers and business partners is severely impaired. In the long term, this can lead to a decline in business relationships and sales (Camillo, 2017). Small and medium-sized enterprises (SMEs) are particularly prone to increasingly frequent and sophisticated cyberattacks (van Haastrecht et al., 2021), as limited financial and human resources often

* Corresponding author.

E-mail address: Thomas.Keller@uni-hohenheim.de (T. Keller).

<https://doi.org/10.1016/j.cose.2026.104952>

Received 10 December 2025; Received in revised form 21 April 2026; Accepted 29 April 2026

Available online 2 May 2026

0167-4048/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

hinder the implementation of comprehensive cybersecurity measures (Johannsen et al., 2020). The absence of dedicated IT security departments further amplifies their vulnerability (Saban et al., 2021). Therefore, employees in SMEs operate in environments with heightened security risks, especially those in common administrative or office-based roles who routinely handle large amounts of sensitive information (Siponen and Vance, 2010).

In response to the increasing number of successful cyberattacks on SMEs and the associated consequences, many companies have introduced information systems security policies (ISSP) (Cram et al., 2017). These are formalized sets of rules and procedures designed to protect an organization's information systems, data, and other digital resources from various threats, including cyberattacks, data breaches, and unauthorized access (Whitman and Mattord, 2018). According to Ifinedo (2014), an ISSP is a formal, explicitly stated organizational policy that specifies clear requirements and rules intended to guard the organization's information systems and related assets against misuse, intentional abuse, or destruction. These policies describe best practices and security measures that employees should follow to ensure the integrity, confidentiality, and availability of sensitive information (Dhillon and Backhouse, 2001; Waqdan et al., 2025).

However, achieving and maintaining high levels of compliance with these policies is not without its challenges (Alotaibi et al., 2016). The constantly evolving nature of cyber threats requires regular updates and revisions to ISSP, which makes it challenging for organizations to keep their employees consistently informed (Siponen and Willison, 2009). The complexity of these policies can also lead to confusion, especially if employees are not adequately trained or if they perceive policies as overly restrictive (Kirlappos et al., 2014). This situation is exacerbated when a lack of knowledge regarding effective strategies to mitigate cybersecurity threats is coupled with a lack of willingness and motivation to apply the appropriate measures (Bouraffa and Hui, 2025). Thus, the subjective perception that following these protocols is inconvenient or time-consuming poses a major obstacle to ISSP compliance (Vance et al., 2012), as it directly reduces the willingness and motivation to adhere to them. For example, employees may bypass vital security measures, such as using complex passwords or enabling two-factor authentication, in favor of simpler, less secure practices that they deem to be more efficient in their operative workflow (Warrington et al., 2021). It is important to note that perceptions of the trade-off between security and convenience vary widely among employees. A common approach to accounting for these differences is to focus on broad personality traits. These constructs describe characteristic patterns of thoughts, feelings, and behaviors that an individual exhibits consistently and stably in multiple situations and interactions (Diener and Lucas, 2026). However, this very defining feature implies that personality traits function as distal influences on decisions and actions in specific situations, such as in the workplace. Therefore, they offer limited insight into the reasons why employees choose to comply with particular rules and procedures at work or why they choose not to.

Against this background, the development of comprehensive ISSP and related training needs to be accompanied by measures that foster employees' compliance by addressing relevant psychological drivers of security behavior. These drivers include relatively stable, person-bound tendencies that are decontextualized and cross-situational in nature (i. e., general personality traits) as well as forms of action regulation that are more proximal to the required behavior and its context (i. e., domain-specific motivational modes). As the extent and mode of motivation influence how employees make decisions and adhere to rules and procedures (Van den Broeck et al., 2021), motivational aspects of regulating secure workplace activities should contribute to ISSP acceptance over and above general personality traits. For example, while a general tendency towards conscientiousness may facilitate ISSP adherence, a sense of commitment to minimizing security risks should play a distinctive role due to its domain-specific focus.

Effective motivational appeals in training and awareness measures

require a thorough understanding of both general and domain-specific drivers of compliance. However, despite the arguably vital function of motivational action regulation, empirical research in cybersecurity remains surprisingly limited. Existing studies often rely on broad attitude- or intention-based instruments rather than on elaborated motivational frameworks and theoretically grounded motivational constructs (Balagopal and Mathew, 2024).

Against this backdrop, the present study aims to explain interindividual differences in ISSP compliance intention by moving beyond trait-only constructs and examining domain-related modes of action regulation as an explanatory layer with close proximity to the behaviors of interest.

1.1. State of research: personality traits and ISSP compliance

The term 'personality' is defined as "psychological qualities that contribute to an individual's enduring and distinctive patterns of feeling, thinking, and behaving" (Jani and Han, 2013, p. 972). In other words, individuals can be distinguished from one another based on their consistent, stable patterns of perceiving and responding to their environment. This is why personality characteristics are often referred to as traits (Fetchenhauer, 2011; Neyer and Asendorpf, 2024). The most commonly used model for describing these traits is the Big Five model (McCrae and John, 1992), which distinguishes five fundamental dimensions of personality. *Extraversion* characterizes individuals as outgoing, energetic, and inclined to enjoy social interaction. High degrees of extraversion make a person more likely to be assertive and communicative and to seek excitement and stimulation. *Agreeableness* reflects a person's tendency to be compassionate, cooperative, and friendly. Highly agreeable people are empathetic, helpful, and value getting along with others. Individuals who exhibit high levels of *conscientiousness* are typically characterized by their meticulous organization, reliability, and a strong inclination towards achieving their objectives. They are disciplined, responsible, and pay attention to detail. High levels of *neuroticism* are associated with emotional instability, anxiety, and moodiness. Markedly, neurotic individuals experience stress, worry, and negative emotions more frequently than their emotionally stable counterparts. *Openness to experience* reflects a person's imagination, creativity, and curiosity. People with high levels of openness are willing to explore new ideas, embrace change, and seek out new experiences.

It is proposed that individual positions along these dimensions influence daily behavior in a multitude of contexts, including social interactions, occupational choices, and compliance with organizational rules (Ozer and Benet-Martínez, 2006). In accordance with this assumption, findings from a recent systematic review by Grandhi and Still (2025) suggest that individual personality traits are associated with the extent of cybersecurity awareness and a range of (predominantly self-reported) cybersecurity behaviors, including security compliance and related practices (e.g., phishing susceptibility and password practices). Conscientiousness, in particular, shows consistent empirical associations with information security-related outcomes. Individuals high in conscientiousness are typically characterized by self-discipline, rule orientation, and a strong sense of responsibility, which explains their demonstrably greater tendency to adhere to security guidelines and organizational requirements (McCormac et al., 2017; Pattinson et al., 2015). Agreeableness has also frequently been linked to higher self-reported security compliance and protective cybersecurity practices, which may be attributed to a stronger orientation towards social norms and a greater consideration of others' expectations, including those of organizations (Schmeisser et al., 2021; Roccas et al., 2002). Openness has been discussed as another relevant trait, showing associations with self-reported cybersecurity awareness and behaviors (Grandhi and Still, 2025). Conceptually, openness indicates a greater tendency to seek, value, and integrate new information (Jach and Smillie, 2020), which may increase engagement with cybersecurity-related content and encourage the adoption of proactive

protective measures. In contrast, extraversion and neuroticism are more frequently connected to self-reported risky or less compliant information security behavior (Shappie et al., 2020). This may be explained by the fact that extraverted individuals are more prone to risk-taking behavior and overconfidence (He and Lei, 2025), which can hinder adherence to security policies. Neuroticism, in turn, has been linked to stress-related mistakes, heightened insecurity, and increased error proneness in security-relevant situations (López-Aguilar et al., 2025; Diamant and Rusou, 2021).

Despite these recurring patterns, extant research is constrained by conceptual and methodological limitations. On the one hand, personality traits are often examined in isolation. This perspective makes it difficult to determine their relative importance in the presence of other traits or psychological factors (Grandhi and Still, 2025). On the other hand, although several Big Five dimensions show statistically significant and positive associations with security-related outcomes, the variance explained by personality traits remains consistently small (Gratian et al., 2018; Warrington et al., 2021). Taken together, these limitations suggest that personality provides at best a modest distal contribution to self-reported security-related compliance behavior, which underscores the need to incorporate more proximal psychological and contextual factors in explanatory models.

1.2. Domain-specific motivational modes for ISSP compliance

Motivation is a central psychological construct that drives individuals to initiate, maintain, and direct their behavior toward specific goals. It encompasses the internal and external forces that influence an individual's decisions and actions in given situations or contexts (Pinder, 2008). Thus, motivation is a dynamic phenomenon shaped by both situational incentives (e.g., rewards or social recognition) and individual dispositions (e.g., needs, preferences, and motives; Heckhausen and Heckhausen, 2018; Scheffer and Heckhausen, 2018). It governs not only the initiation of actions and the formation of goals, but also the persistence and vigor with which individuals engage in targeted behaviors. In contrast to general tendencies of thinking, feeling, and (re-)acting, the intensity and quality of an individual's motivation may vary depending on the context and task. Thus, "being motivated" to perform certain activities, such as fulfilling a particular set of work duties or participating in a type of sport, denotes a domain-specific construct (Bjerg Hall-Andersen and Broberg, 2014; Michel et al., 2022).

In the domain of workplace cybersecurity, Furnell and Rajendran (2012) suggest that professionals' ISSP-compliant behavior may vary along a continuum, ranging from commitment to an accepted security culture at one end to indifference and disobedience at the other. Related empirical research mostly draws on models from social psychological theories. The general deterrence theory (GDT) posits that the fear of sanctions and penalties can prevent employees from deviating from organizational regulations (D'Arcy et al., 2009; Siponen and Vance, 2010; Workman et al., 2009). Conversely, the protection motivation theory (PMT; Rogers, 1975) proposes that individuals are more inclined to engage in ISSP-compliant behaviors when they perceive the need for protecting organizational assets (threat assessment) and believe that they are capable of doing so (self-efficacy). While GDT and PMT place different emphases on externally and internally motivating factors, empirical evidence lends support for both models (GDT: Trang and Brendel, 2019; PMT: Boss et al., 2015; Johnston and Warkentin, 2010). This suggests that ISSP-compliant workplace behaviors (as well as their absence) can have different or even multiple causes of action, demanding an integrative framework for analysis.

The renowned self-determination theory of motivation (SDT; Deci and Ryan, 1985) offers such an integrative perspective. It assumes autonomous and controlled modes of action regulation to be guiding an individual's decisions and actions to varying degrees. Thus, SDT elaborates the specific *modes of motivation*, thereby indicating the salient degree of self-regulation involved in an individual's choice and course of

action, as shown in Fig. 1. The value of employing SDT as a theoretical foundation stems from its nuanced perspective on the underlying factors that motivate workplace behaviors and their consequences, as will be outlined below with a focus on ISSP-compliant practices. Another advantage is its well-established assumptions about interventions that facilitate progression from externally to internally regulated activities, a topic that will be addressed in Section 4.3.

Intrinsic motivation denotes the purely internal drive to engage in a behavior or task for its own sake, i.e., its intrinsic value. Action steps are regulated primarily by the inherent satisfaction they provide and, thus, by an individual's perception of the activity or tasks as being fulfilling, meaningful, interesting, and genuinely enjoyable. Those who are intrinsically motivated to comply with security policies do so because they have a genuine understanding of and appreciation for the importance of protecting the organization's assets. Furthermore, they find the protection of data and IT systems personally satisfying, interesting, and exciting. SDT distinguishes *four categories of extrinsic motivation*. These categories differ in their degree of perceived autonomy (locus of action control) and the extent to which behavioral goals are internalized. Accordingly, *external regulation* represents the most externally controlled mode of behavior, wherein actions are driven exclusively by rewards or punishments stemming from outer sources. For example, employees may adhere to safety policies solely to circumvent disciplinary measures or receive recognition. In instances of *introjected regulation*, an individual has begun to internalize formerly external incentives or pressures. Compliance might be motivated by internalized guilt or anxiety, such as the fear of being blamed for a security breach. In the *identified regulatory* stage, the person acknowledges and accepts the necessity or function of the behavior in question. Employees may follow ISSP because they recognize their relevance for organizational safety. *Integrated regulation* is the most autonomous form of extrinsic motivation, in which activities are fully and harmoniously integrated into an individual's system of values and goals. On the other side of this spectrum lies *amotivation*, demarcating a state of absent motivation. According to the SDT classification, this is the only state in which *any* directing and regulating influences on behavior from either internal or external sources are missing. Amotivated employees ignore ISSP because they are neither convinced of their usefulness, nor committed to them as an integral part of their professional ethos, nor enthusiastic about their mastery, nor are they prompted to implement them by impending sanctions. This lack of motivation may stem from feelings of incompetence, a disconnect with organizational goals, or the belief that their actions will not make a difference.

A meta-analysis (Van den Broeck et al., 2021) shows that intrinsic regulation is the strongest predictor of employee well-being, attitudes, and behavior. However, identified regulation proves to be a stronger predictor of performance and organizational citizenship behavior. This suggests that while intrinsic regulation enhances overall engagement and satisfaction, identified regulation is more closely linked to key outcomes like job performance and proactive behavior in organizational contexts because employees adopt and accept organizational goals and values.

Despite these insights, research on the motivational underpinnings of ISSP-compliant behavior often relies on a rather coarse distinction between intrinsic and extrinsic motivation, equating the latter with the external regulation of workplace procedures (Herath and Rao, 2009; Kranz and Häusinger, 2014). From an SDT perspective, common compliance mechanisms such as sanctions and surveillance are typically associated with controlled regulation (Padayachee, 2012). Conversely, beliefs about personal agency and effectiveness, such as autonomy and self-efficacy, are generally treated as markers of more autonomous regulation (Talib, 2015). By comparison, motivational modes that reflect internalization, i.e., the extent to which organizational goals are self-endorsed, have received relatively little attention. Other studies in the broader workplace motivation literature show that 10–25% of employees are amotivated, highlighting both the importance of different

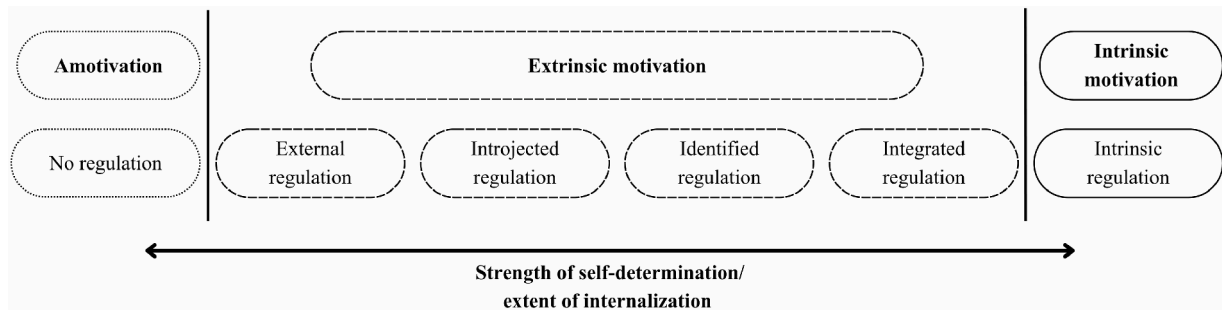


Fig. 1. Continuum of controlled and autonomous motivation for (workplace) behaviors (adapted from Ryan and Deci, 2020).

motivational modes and the phenomenon of lacking behavioral incentives and regulation from any source (Howard et al., 2016).

The SDT's model assumptions on continuously graduated modes of action regulation also have implications for the analytical strategy (Deci et al., 2017; Morin et al., 2020; Tóth-Király et al., 2021). While high amotivation markedly reduces all other motivational modes, this is not necessarily the case among the remaining intrinsic and extrinsic qualities. These qualities can be salient to varying degrees and in different combinations within an individual, even though there must always be a dominant mode of action regulation. For instance, it is possible that an employee carries out a professional task primarily because they perceive it as a highly accepted and thus almost self-imposed duty, but also because they derive some enjoyment from performing it. In this case, task motivation would be mostly introjected but also intrinsic to a comparably lesser degree. Therefore, it is crucial to uncover distinct profiles of domain-related motivational modes among the workforce to get a nuanced picture of why and how different groups of employees are or can be further encouraged to implement certain procedures (Howard et al., 2016; Gillet et al., 2020). This person-centered approach overcomes the limitations of prominent one-dimensional and multi-dimensional approaches. The former fail to account for the unique quality of each mode of action regulation by imposing a one-factor structure on work- or task-related motivation, while the latter cannot estimate a person's position on the self-determination continuum (Deci et al., 2017; Tóth-Király et al., 2021). Recent qualitative evidence suggests that employees may have different motivational rationales for security-related practices, and these rationales may even vary depending on the type of practice (Frank and Kohn, 2023). These findings indicate the presence of different regulatory modes and provide further evidence in support of a person-centered perspective on motivational profiles.

1.3. The present study

The present study examines how both stable individual differences (e.g., Gratian et al., 2018) and domain-specific motivational regulation of behaviors (e.g., Chen et al., 2022) affect ISSP compliance intention. The former are rooted in personality traits that shape employees' general tendencies to think, feel, and act across different tasks and situations (Fetchnauer, 2011). The latter reveals how employees regulate professional activities that should align with their company's ISSP.

In light of the reviewed evidence on the differing *motivational underpinnings* of workplace behaviors and the *personality dependence* of cybersecurity behavior, it seems imperative to include the full spectrum of potential predictors when examining professionals' acceptance and adoption of their organization's security standards. Current research on ISSP compliance intention, focusing specifically on motivation, emphasizes the pivotal role of intrinsic motivation while overlooking significant variations in extrinsic motivation and the concept of amotivation. Thus far, it has not utilized a nuanced configurative approach.

As depicted in Fig. 2, the present study aims, in a theory-informed exploratory manner, to clarify the relative contributions of both domain-specific (motivational) factors and general (personality-bound) tendencies in predicting compliance intention. To this end, the study draws on self-determination theory (SDT; Deci and Ryan, 1993) and the five-factor model of personality (FFM; John et al., 1991). By investigating office-based employees in SMEs, it concentrates on a widespread professional context that is particularly vulnerable to security threats (Rombaldo et al., 2023). Typically, office-based employees in SMEs handle sensitive data without access to dedicated IT security support or comprehensive training (Siponen and Vance, 2010).

Thus, the study addresses the following exploratory research question: *How do (a) personality traits and (b) distinct motivational profiles*

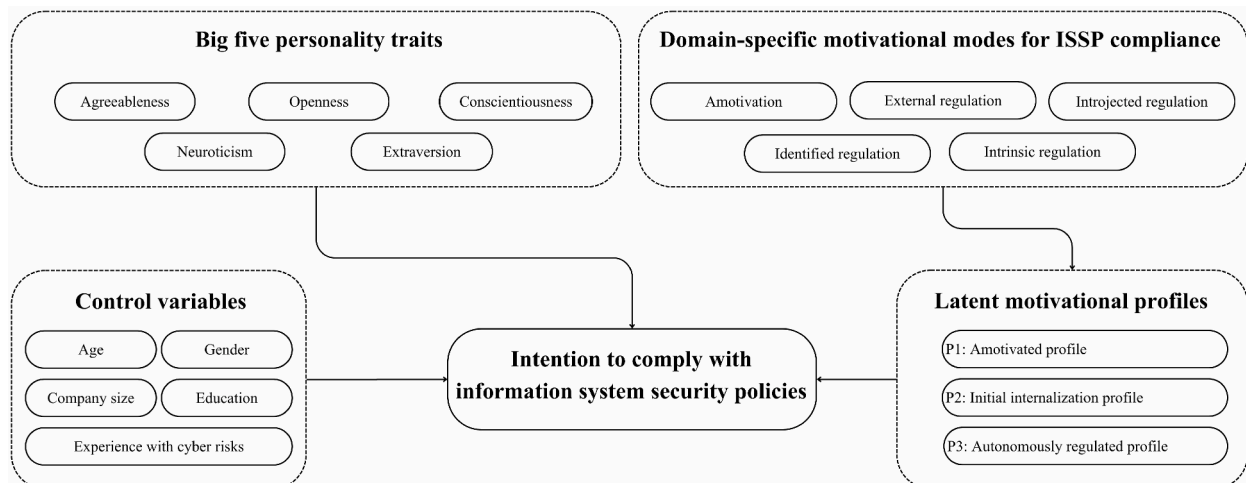


Fig. 2. Conceptual research model linking personality traits and latent motivational profiles to ISSP compliance intention.

relate to professionals' compliance intention with their organization's ISSP in office-based work contexts? By expanding the analytical focus to include the motivational modes that underlie the acceptance of cybersecurity regulations, this research offers valuable insights for organizations seeking to foster long-term, voluntary compliance (Profeti and Toth, 2025).

2. Method

2.1. Participants

To address this research question, we conducted a survey in June 2025 via the online panel provider Prolific among employees of SMEs with the common professional profile of office clerk. Prolific's pre-screening features enabled broad yet targeted recruitment by occupational profile and company size, ensuring close alignment with the intended office-work context in SMEs while reducing gatekeeper-related selection effects (Brodeur et al., 2025). Using this platform also supports the collection of informative and reliable survey responses, as evidenced by prior platform comparisons on key quality indicators such as attention, instruction comprehension, honesty, and scale reliability (Peer et al., 2022).

Table 1 presents the demographic characteristics of the sample, which consisted of N = 218 participants. Of these, 37.6% were female and 62.4% were male. Participants' ages ranged from 19 to 65 years, with a mean age of 32.48 years (SD = 9.15). The largest age group was 31 to 40 years (36.7%), followed by those aged 25 to 30 (30.3%) and 18 to 24 (17.9%). In terms of educational level, 61.5% of the sample held a university degree. Regarding company size, more than half of the participants (55.5%) worked in organizations with 50 to 249 employees, 33.5% in companies with 10 to 49 employees, and 11.0% in companies with nine or fewer employees. Most respondents (88.5%) reported having prior experience with cyber risks.

2.2. Materials and measures

To analyze different modes of action regulation, we used the Multi-dimensional Work Motivation Scale (MWMS; Gagné et al., 2015), which we adapted to the context of cybersecurity behavior. Consequently, all items pertain to participants' endeavors to protect sensitive data and information from unauthorized access, loss, theft, or damage. The sub-scales include *amotivation* (e.g., "I don't make an effort because I feel like I'm wasting my time with this work"), *external regulation* (e.g., "I do make an effort to receive recognition from my superiors and colleagues"),

introjected regulation (e.g., "I do make an effort because I have to prove to myself that I'm capable of it"), *identified regulation* (e.g., "I do make an effort because it is important to me personally"), and *intrinsic regulation* (e.g., "I do make an effort because I enjoy it") of ISSP-compliant workplace behaviors. Personality traits were assessed using the Big Five Inventory (BFI-44), which was developed by John and Srivastava (1999). This 44-item self-report questionnaire measures five broad personality dimensions: extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience. ISSP compliance intention was assessed using the ISSP compliance behavioral intentions scale (Ifinedo, 2014), which captures employees' intentions to adhere to organizational ISSP and is widely used in cybersecurity research. The questionnaire was translated into German by independent bilingual experts following standardized guidelines for cross-cultural instrument adaptation (Beaton et al., 2000).

Participants answered all items of the employed scales on a 6-point Likert-type scale (from 1 = does not apply at all to 6 = applies completely). Additionally, they reported age, gender (coded 0 = female, 1 = male), highest educational level (coded 0 = no university degree, 1 = university degree), experience with cyber risks (coded 0 = no experience, 1 = experience), and company size as control variables.

2.3. Data preparation

To ensure robust and interpretable results, we applied a multi-stage data cleaning procedure that considered both data quality and theoretical consistency. First, we checked the responses for formal data integrity. Participants who failed predefined attention checks (n = 9) or completed the survey in less than 50% of the time allotted (n = 31) were excluded at this stage (Meade and Craig, 2012).

Next, we addressed theoretical consistency within the motivational data. Based on SDT's assumptions, laid out in Section 1.2, we identified and removed cases exhibiting incongruent answers. Specifically, we calculated an intra-individual response variability (IRV) index to assess within-person variability across each participant's self-reported modes of action regulation (Dunn et al., 2018). Cases with very low variability, indicating undifferentiated response patterns across all motivational scales, were excluded as theoretically inconsistent with assumptions on the relative prevalence of one mode (n = 9).

To identify extreme response patterns, we examined the dataset for univariate outliers using standardized z-scores and boxplot visualizations, as well as for multivariate outliers within the sample using Mahalanobis distance (Hair et al., 2019). While the analysis did not identify any multivariate outliers, a few univariate extreme values emerged in individual variables. These cases were not removed from the dataset, as they reflected plausible individual differences rather than data entry errors or atypical response patterns. Moreover, their number was too small to exert a disproportionate influence on the results.

Because both the explanatory variables (personality, motivation) and the outcome (ISSP compliance intention) were assessed via self-report in the same survey, we took several procedural steps to mitigate potential common method variance (Conway and Lance, 2010). Participation was fully anonymous via Prolific, instructions emphasized that there were no right or wrong answers, and scale blocks were presented in randomized order to reduce consistency motifs and order effects (Podsakoff et al., 2003). Additionally, we included an unrelated marker construct (Miller and Simmering, 2023) to probe for shared method variance. The marker exhibited only negligible associations with the focal constructs. When entered as an additional covariate in the full regression model, it failed to predict compliance intention (b = 0.04, p = .345) or enhance model fit. Although common method effects can never be fully ruled out in a cross-sectional self-report design, the checks we performed suggest that these effects are unlikely to account for the observed relationships in the present dataset.

Following the data check and cleaning process, the final sample of 218 participants was used for the main analyses.

Table 1
Sociodemographic characteristics of participants (N = 218).

	n	%
Gender		
Female	82	37.6
Male	136	62.4
Age		
18 - 24	39	17.9
25 - 30	66	30.3
31 - 40	80	36.7
41 - 50	21	9.6
51 - 60	9	4.1
> 60	3	1.4
Education		
No university degree	84	38.5
University degree	134	61.5
Experience with cyber risks		
Yes	193	88.5
No	25	11.5
Company size		
≤9	24	11
10-49	73	33.5
50-249	121	55.5

2.4. Statistical analyses

2.4.1. Measurement model validation

To examine the construct validity of the MWMS, we conducted a confirmatory factor analysis (CFA) to determine if the empirical structure of the survey data aligned with the SDT framework. This step was necessary to validate the adapted scale in a cybersecurity context. We estimated the model using the robust maximum likelihood estimator (MLR) to compensate for potential deviations from multivariate normality. Model fit was assessed using the following established fit indices: the Comparative Fit Index (CFI), the Tucker-Lewis Index (TLI), the Root Mean Square Error of Approximation (RMSEA), and the Standardized Root Mean Square Residual (SRMR). Following the recommendations of [Hu and Bentler \(1999\)](#), we interpreted CFI and TLI values of 0.95 or higher, an RMSEA of 0.06 or lower, and an SRMR of 0.08 or lower as indicative of good model fit. In contrast, no CFA was performed for the personality traits because their factor structure has been extensively validated in prior research. The reliability of the constructs was evaluated using Cronbach's α , with values of 0.70 or higher indicating acceptable internal consistency ([Nunnally and Bernstein, 1994](#)).

2.4.2. Latent profile analysis

We ran a latent profile analysis (LPA) with standardized factor scores derived from the CFA of the MWMS. The LPA, estimated with MLR, aimed to identify homogeneous subgroups of individuals with distinct motivational profiles, maximizing similarity within profiles and distinctiveness between them ([Lubke and Muthén, 2005](#)). We examined competing models with one to five profiles under the assumptions of equal variances across profiles and no covariances among indicators. Model selection was informed by a combination of statistical fit indices, classification quality, and theoretical interpretability ([Nylund-Gibson and Choi, 2018](#)). Specifically, the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the sample size adjusted BIC served to evaluate model fit, with lower values indicating a better fitting model considering model complexity. In addition, Bootstrap Likelihood Ratio Tests (BLRT) were used to compare models with k versus $k-1$ profiles, with significant results indicating that the model with the larger number of profiles provided a statistically superior fit ([Bauer, 2022](#)). Classification precision was assessed using entropy, which ranges from 0 to 1. Values above 0.80 are considered indicative of high classification accuracy ([Muthén, 2004](#)), while lower values suggest less distinct profile separation. Additionally, average posterior classification probabilities were examined to evaluate the certainty of profile assignment. Values approaching 1.00 indicate a high likelihood that individuals were correctly classified ([Jung and Wickrama, 2008](#)). Profile sizes were also considered because very small classes (e.g., fewer than 5% of the sample) may be unstable and less generalizable ([Weller et al., 2020](#)). To demonstrate the robustness and validity of the selected profile solution, we conducted an additional split-half cross-validation procedure ([Masyn, 2013](#)). To this end, the full sample was randomly divided into two subsamples of equal size. Latent profile analyses were then performed independently in both subsamples while employing the same model specifications and selection criteria. Model fit indices (AIC, BIC, and sample-size adjusted BIC), entropy, and minimum class size were evaluated separately in each subsample to examine whether the same number of profiles emerged. In addition, the substantive similarity of the profile solutions across subsamples was assessed by comparing the mean levels of the action regulations within each profile. Profile alignment procedures were applied to address potential label switching, enabling a direct comparison of profile shapes across subsamples.

2.4.3. Hierarchical regression analysis

Finally, we conducted a hierarchical regression analysis to examine the relationships between the assumed predictors and ISSP compliance intention. First, we calculated means, standard deviations, and Pearson correlations for all scales and control variables to establish a descriptive

baseline and identify empirically relevant predictors. We also assessed the potential for multicollinearity among the predictors by calculating Variance Inflation Factors (VIF). All VIF values were below 3, suggesting that multicollinearity did not pose a significant issue ([Hair et al., 2011](#)). We created an initial regression model that included demographic and structural characteristics to account for their share of the variance in employees' compliance intentions. The second model incorporated personality traits to assess their relative contribution to predicting compliance intention. A third model added motivational profiles, derived from the latent profile analysis of the MWMS, as categorical predictors. This sequence reflects the theoretical assumption that motivational profiles specific to the domain of information security possess greater proximity to decisions and actions than structural/demographic and personality characteristics. This approach also aligns with the study's objective of investigating whether motivational profiles can explain variance in compliance intention beyond established personality traits.

All analyses were performed using R version 4.3.2, employing the lavaan package for structural equation modeling and the tidyLPA package for LPA ([Rosenberg et al., 2018](#); [Rosseel, 2012](#)).

3. Results

3.1. Measurement model selection and preliminary results

Prior to regression analyses, the underlying measurement structure of motivational regulation of cybersecure behavior was examined to ensure an adequate representation of the theoretical constructs. Several competing CFA models with increasing degrees of differentiation between regulation modes were compared (see [Appendix A](#)). Overall, the results supported a five-factor structure consistent with SDT, with additional subdimensions for external and introjected regulation (see [Fig. 3](#)). This model provided the best balance between theoretical coherence and empirical fit. In the selected measurement model, external and introjected regulation were specified as second-order factors to account for distinct semantic aspects of the survey items. Specifically, external regulation was represented by two first-order subdimensions that captured material incentives and social pressure related to engaging in information-secure behavior at work, respectively. Introjected regulation was split into approach-oriented and avoidance-oriented components. In this hierarchical specification, the subdimensions are measured directly by the items, while the second-order factors represent the shared regulatory core of each construct. This refined structure yielded the best fit of all the tested models (TLI = 0.934, CFI = 0.947, RMSEA = 0.065, and SRMR = 0.064). Its standardized factor loadings support both the theoretical structure of the SDT framework and the structure of the adapted domain-specific work motivation scale (MWMS). In particular, all items significantly loaded on their respective first-order factors, exhibiting moderate to strong standardized loadings above 0.50. The second-order factors also showed substantial loadings on their lower-order components.

Subsequently, descriptive statistics and bivariate correlations were examined to provide an initial overview of all variables. The descriptive statistics indicate appropriate means and standard deviations across all variables, accompanied by good to excellent internal consistencies (see [Table 2](#)). Amotivation ($\alpha = 0.87$), identified regulation ($\alpha = 0.87$), and intrinsic motivation ($\alpha = 0.90$) show particularly high reliability. External regulation ($\alpha = 0.81$) and introjected regulation ($\alpha = 0.77$) also demonstrate solid reliability. With an alpha of 0.92, the ISSP compliance intention scale indicates highly consistent responses across its items. Likewise, the measure of the Big Five personality traits proved to be solidly reliable. Extraversion ($\alpha = 0.86$), conscientiousness ($\alpha = 0.86$), neuroticism ($\alpha = 0.86$), and openness ($\alpha = 0.82$) show good to very good consistency. Agreeableness still meets accepted thresholds for adequate internal consistency ($\alpha = 0.74$). Correlation patterns are theoretically coherent. Conscientiousness, agreeableness, and openness show the

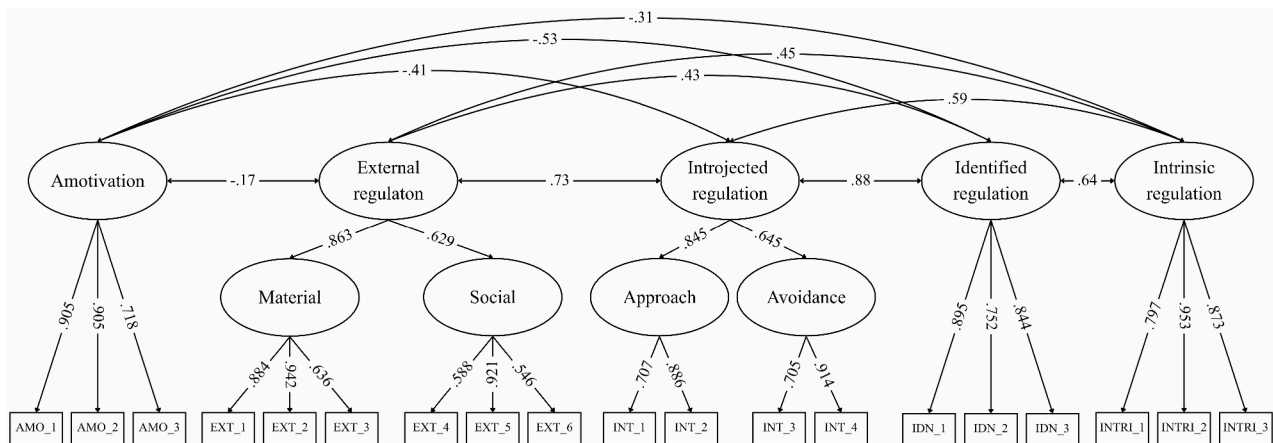


Fig. 3. Final hierarchical MWMS measurement model with second-order external and introjected regulation factors.

strongest positive associations with ISSP compliance intention, whereas extraversion is only weakly and non-significantly related to the target variable. Neuroticism is negatively related to all other traits and shows a small negative correlation with ISSP compliance intention. The motivational variables exhibit theoretically coherent relations with the target variable. Specifically, identified, intrinsic, introjected, and external regulation show positive relationships with ISSP compliance intention. Amotivation, by contrast, is negatively linked to all relevant variables. Moreover, the intercorrelations among the MWMS scales follow the expected SDT continuum pattern: conceptually adjacent modes of regulation (e.g., external–introjected, identified–intrinsic) correlate more strongly than conceptually distant regulations (e.g., external–intrinsic). In sum, the descriptive results indicate that the scales are functioning well and provide initial evidence that both motivational and dispositional factors can play meaningful roles in shaping ISSP compliance intention.

3.2. Identification of motivational profiles in cybersecurity behavior

The LPA, based on the five motivational regulation modes, revealed notable differences in model fit across the tested solutions with one to five profiles (see Table 3). As expected, the fit indices improved continuously with increasing numbers of profiles. From a purely statistical perspective, this pattern favors more complex models. In addition, the BLRT value was significant for all successive model comparisons (all $p < .01$), indicating improved fit with each additional profile. However, the final number of profiles was determined by balancing global fit against classification quality, class size distribution, and substantive interpretability. The one-profile solution served as a baseline, showing perfect entropy (1.00) by definition. The two-profile solution showed acceptable class separation (entropy = 0.775) and balanced class sizes (minimum class = 45%), but it provided an overly coarse representation of motivational heterogeneity. The three-profile solution achieved a substantial improvement in fit (BIC = 2626; SABIC = 2557) and very good classification accuracy (entropy = 0.881), with the smallest class comprising 8% of the participants and a minimum posterior probability of 0.923. While the four- and five-profile solutions yielded only marginal additional improvements in information criteria, they were accompanied by reduced class separation (entropy = 0.789 for four profiles) and smaller classes (minimum class = 6% in the five-profile solution). Moreover, the split-half cross-validation analysis confirmed the three-profile solution as the most stable model with respect to sampling variability (see Appendix B, Table B1 and B2). Taken together, these results support retaining the three-profile solution as the final model.

Based on the selected solution, three distinct and conceptually interpretable motivational profiles emerged for the domain of cybersecurity behavior in office workplaces (see Fig. 4). These profiles align

with the continuum proposed by SDT, ranging from amotivation to controlled and autonomous modes of regulation that manifest at different intensity levels. Thus, they reflect qualitative and quantitative differences in motivational functioning. The first and smallest profile can be labeled as *amotivated*. It comprises approximately 8% of employees who display an indifferent neglect towards cybersecurity rules and practices. This profile shows the highest level of amotivation among all profiles, combined with low expressions of both controlled and autonomous modes of motivation. This pattern indicates a profound lack of intentional regulation, which is typically associated with low levels of subjective competence, limited personal relevance, or a sense of inefficacy (Ryan and Deci, 2017). Employees with this profile appear disengaged from security goals and guidelines and consider secure workplace behaviors to be neither self-directed nor mandatory. The third profile contrasts sharply with the first due to its *high level of autonomy* in regulating cybersecure behavior. It accounts for about 39% of employees who demonstrate a highly internalized and, thus, self-determined mode of motivation. This group shows the lowest level of amotivation and the highest levels of identified and intrinsic regulation, accompanied by above-average controlled regulation (in particular, the introjected mode). This configuration reflects a high commitment to and a deep understanding of the importance of cybersecure behavior. This behavior is perceived as personally meaningful, valuable, and aligned with one's goals. Within SDT, such a pattern is considered high-quality, volitional motivation, typically associated with sustained engagement, consistent enactment, and adaptive coping (Deci and Ryan, 2000). The second and largest profile, named *initial internalization*, encompasses roughly 53% of the sample and reflects a motivational pattern of emerging internalization. In other words, employees with this motivational profile demonstrate a heightened, albeit not yet firmly established, willingness to engage in cybersecure behavior. Apparently, they have begun to recognize the relevance of such practices and are in an early stage of integrating them into their understanding of their professional responsibilities. With introjected and identified modes of action regulation being more prominent than in the first profile, though still falling below the levels of the third profile, this pattern delineates a comparably diffuse motivational state. From an SDT perspective, it reflects a nascent stage of internalized cybersecure behavior in which self-regulatory processes are present but still relatively weak.

Taken together, the three profiles reveal substantial heterogeneity in employees' motivation to adopt cybersecure behaviors. In subsequent hierarchical regression models, they serve as categorical predictors, enabling us to examine their incremental contribution to predicting ISSP compliance intention over and above empirically established personality factors.

Table 2
Means, standard deviations, correlations, and internal consistencies (Cronbach's alpha).

Variable	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Age	32.48	9.15																
2. Gender (1 = Male)	0.62	0.49	−0.00															
3. Education (1 = University degree)	0.61	0.49	.15*	.03														
4. Company size	2.44	0.69	.14*	.05	.21**													
5. Experience with cyber risks (1 = Yes)	1.11	0.32	−0.07	−0.02	−0.13	−0.15*												
6. Extraversion	3.44	0.90	.17*	−0.03	.03	.06	.00	(.86)										
7. Agreeableness	4.40	0.63	.04	.02	−0.01	.11	.01	.23**	(.74)									
8. Conscientiousness	4.35	0.79	.27**	−0.06	.10	.03	.05	.41**	.32**	(.86)								
9. Neuroticism	3.23	0.89	−0.16*	−0.28**	−0.18**	−0.10	−0.01	−0.49**	−0.34**	−0.42**	(.86)							
10. Openness	4.20	0.77	.08	−0.14*	−0.08	.02	−0.15*	.29**	.16*	.25**	−0.13	(.82)						
11. Amotivation	1.93	0.91	−0.08	−0.15*	.17*	.13*	−0.03	−0.09	−0.20**	−0.32**	.13	−0.18**	(.87)					
12. External regulation	3.61	0.97	−0.07	−0.07	−0.08	−0.01	.04	−0.03	−0.01	.07	.10	.11	−0.13	(.81)				
13. Introjected regulation	3.99	1.01	−0.06	−0.16*	−0.01	.04	.06	.05	.09	.15*	.02	.15*	−0.26**	.48**	(.77)			
14. Identified regulation	4.45	1.03	.13	−0.13*	−0.04	−0.03	−0.02	.19**	.19**	.31**	−0.15*	.23**	−0.44**	.32**	.64**	(.87)		
15. Intrinsic motivation	3.89	1.21	.09	−0.05	−0.16*	−0.07	−0.02	.23**	.09	.24**	−0.18**	.21**	−0.27**	.31**	.40**	.59**	(.90)	
16. ISSP compliance intention	5.04	0.78	.04	.02	−0.06	.22*	−0.04	.10	.31**	.36**	−0.13*	.23**	−0.45**	.20**	.34**	.35**	.21**	(.92)

Note. M and SD are used to represent mean and standard deviation, respectively. Reliabilities (Cronbach's alpha) are shown in parentheses on the diagonal. * indicates $p < .05$; ** indicates $p < .01$. All items were measured on a 1–6 Likert scale (1 = strongly disagree, 6 = strongly agree).

3.3. Hierarchical multiple regression analysis of ISSP compliance intention

In the first step of the hierarchical regression, the demographic and structural predictors age, gender, education, company size, and experience with cyber risks were entered into the model. This set of predictors explained a small but statistically significant proportion of the variance in ISSP compliance intention, $R^2 = 0.060$. Company size emerged as the only significant factor ($b = 0.27$, $p < .001$), with employees in larger organizations reporting higher levels of ISSP compliance intention. This pattern suggests that the presence of more formalized security structures, clearer responsibilities, and stronger emphasis on policy adherence in larger companies, in comparison to small companies, may foster more compliant conduct.

In the second step, we added the personality traits of extraversion, agreeableness, conscientiousness, openness, and neuroticism. Including these variables led to a substantial and statistically significant increase in explained variance, $\Delta R^2 = 0.197$, $F(5, 207) = 12.12$, $p < .001$. Agreeableness ($b = 0.24$, $p = .003$) and conscientiousness ($b = 0.35$, $p < .001$) emerged as robust positive predictors of ISSP compliance intention. Employees who described themselves as more cooperative, considerate, diligent, and reliable reported a stronger intent to follow their company's ISSP. Openness was also positively related to compliance intention ($b = 0.14$, $p = .035$), suggesting that staff who are open to new experiences and ideas are more willing to adhere to ISSP. However, neither extraversion ($b = -0.08$, $p = .195$) nor neuroticism ($b = 0.05$, $p = .446$) was linked to compliance intention. Importantly, company size remained a significant predictor in this extended model, indicating that certain structural conditions and personality-based dispositions independently account for variance in the target variable.

In the third and final step, we included dummy-coded motivational profiles, using the amotivated profile as the reference category. Adding these profiles yielded a further significant increase in explained variance, $\Delta R^2 = 0.08$, $F(2, 205) = 11.56$, $p < .001$, resulting in a final R^2 of 0.33. Membership in both the initial internalization profile ($b = 0.64$, $p < .001$) and the autonomously regulated profile ($b = 0.87$, $p < .001$) significantly and positively predicted ISSP compliance intention compared to the amotivated profile. Thus, employees who have developed greater awareness and appreciation for cybersecurity-related practices are more willing to integrate them into their professional routines in accordance with their company's ISSP. This pattern contrasts with employees who respond to neither external nor internal incentives or sanctions (amotivated profile) and is most pronounced for those individuals who have internalized ideas of fighting security threats to the comparably greatest extent (autonomously regulated profile). Notably, the predictive value of openness is missing in the final model ($b = 0.10$, $p = .115$), indicating that openness does not play a unique role in explaining ISSP compliance intention beyond motivational profiles and structural conditions. Thus, the previously observed openness effect may partly reflect variance shared with motivational profile membership rather than an independent contribution. All other significant predictors, as well as company size as a relevant covariate, remained stable across the estimated models (see Table 4).

4. Discussion

4.1. Principal findings and theoretical contributions

This study examined how *stable personality traits* and *domain-specific motivational modes* together influence professionals' intentions to comply with their company's ISSP in office settings. Using self-determination theory and the five-factor model of personality, we analyzed 218 office employees from small and medium-sized enterprises in Germany who regularly handle sensitive data. Using construct validation, person-centered motivational profiling, and hierarchical regression, our study provides an integrative account of why employees differ in their ISSP

Table 3

Latent profile analysis results.

No. of Profiles	AIC	BIC	Adjusted BIC	Entropy	Min. profile n	Min. probability	p(BLRT)
1	3005	3039	3007	1.000	218 (100%)	1.000	< 0.01
2	2734	2788	2737	0.775	99 (45%)	0.931	< 0.01
3	2552	2626	2557	0.881	17 (8%)	0.923	< 0.01
4	2535	2630	2541	0.789	17 (8%)	0.807	< 0.01
5	2484	2599	2491	0.831	13 (6%)	0.802	< 0.01

Note. AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; Adjusted BIC = Sample-size adjusted Bayesian Information Criterion (SABIC); Entropy = classification accuracy index; Min. Profile n = size of the smallest class in absolute numbers and percentage of the total sample; Min. probability = average posterior probability for the smallest class; BLRT = Bootstrap Likelihood Ratio Test.

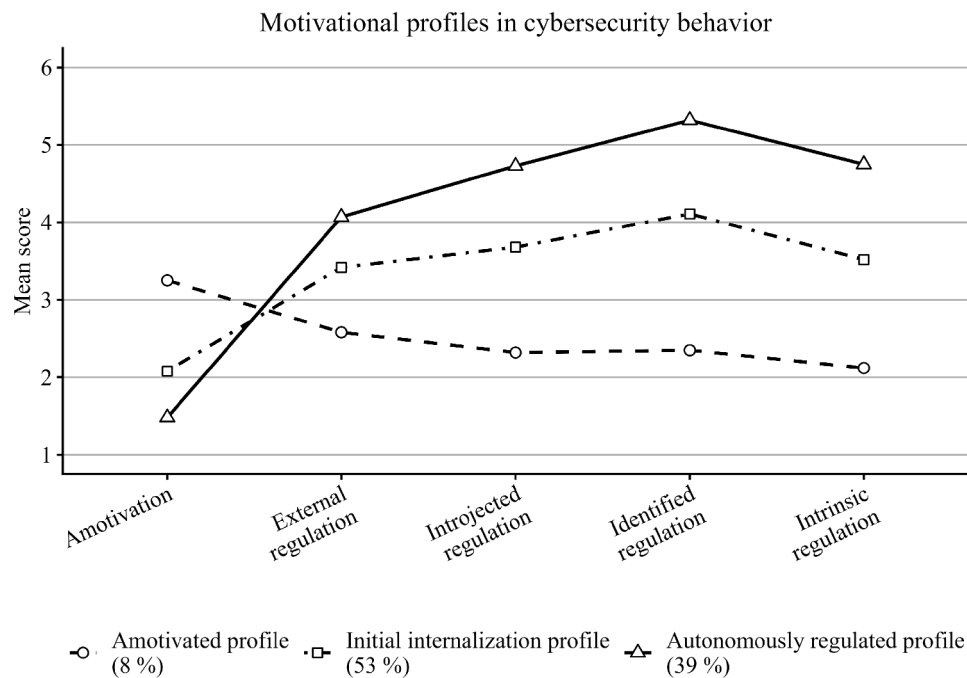


Fig. 4. Motivational regulation patterns for amotivated, initial internalization, and autonomous profiles.

compliance intention.

The *Cybersecurity-MWMS* (see [Appendix C](#)), a nuanced measure of work motivation adapted to the domain of cybersecure behavior at work, demonstrated strong conceptual alignment and an acceptable empirical fit. A refined measurement model supported the expected differentiation of regulation types in line with SDT. However, external and introjected regulation were better represented as hierarchical constructs, with two subdimensions each. These were specified as first-order factors: material-external versus social-external regulation and approach- versus avoidance-based introjection. The full model structure was supported by sufficiently high factor loadings, an intercorrelation pattern consistent with SDT's continuum assumptions, and decent internal consistencies of all subscales. These psychometric properties closely resemble those reported in previous SDT-based validation studies ([Gagné et al., 2015](#)), highlighting the robustness and generalizability of the domain-specific, multidimensional framework. In essence, the adaptation of the MWMS to the domain of cybersecure behavior at work advances research by moving beyond the intrinsic-extrinsic dichotomy and by enabling a theoretically grounded, fine-grained characterization of domain-specific motivational regulation.

A closer examination of the LPA results revealed *three motivational profiles* among office clerks: an amotivated profile, an initial internalization profile, and an autonomously regulated profile. These profiles exhibit systematic differences in their configurations of controlled and autonomous regulation modes, as well as in their levels of amotivation. Within the sample, only 39% of employees exhibited a high level of

autonomous regulation, characterized by strong identified and intrinsic regulation and, thus, by a genuine internalization and valuation of cybersecure behavior at work. In contrast, a small but theoretically critical subgroup of approximately eight percent belonged to the amotivated profile. For these employees, neither external drivers (e.g., sanctions or incentives) nor internal motives (such as perceived value, self-obligation, competence beliefs, or meaningfulness) appeared to be salient. From an SDT perspective, this motivational void renders consistent cybersecure behavior unlikely, as ISSP-related actions are neither externally enforced nor volitionally endorsed. The remaining and largest group, comprising approximately 53% of employees, reflects an initial stage of internalization, characterized by elevated but not yet consolidated identified and introjected regulation. These findings not only attest to substantial heterogeneity in how employees regulate their cybersecurity behavior, but they are also compatible with SDT-based profiling research on general work motivation, which has repeatedly documented substantial variations in motivational patterns ([Gillet et al., 2020](#); [Tóth-Király et al., 2021](#)). By identifying varying regulation modes of cybersecure workplace practices, the present study contributes to existing cybersecurity research. It provides the first evidence of distinct and, in particular, theoretically plausible motivational profiles underlying security-compliant workplace behavior.

Turning to the regression models, our results suggest that *structural, dispositional, and motivational factors* jointly explain variations in employees' ISSP compliance intention. *Company size* and two personality traits – *conscientiousness and agreeableness* – stood out as significant and

Table 4

Hierarchical regression results using ISSP compliance intention as the criterion.

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]	<i>beta</i>	<i>beta</i> 95% CI [LL, UL]	<i>sr</i> ²	<i>sr</i> ² 95% CI [LL, UL]	<i>r</i>	Fit
(Intercept)	4.48**	[3.74, 5.22]						
Age	0.00	[−0.01, 0.01]	0.03	[−0.11, 0.16]	.00	[−0.01, 0.01]	.04	
Gender	0.01	[−0.20, 0.22]	0.01	[−0.12, 0.14]	.00	[−0.00, 0.00]	.02	
Education	−0.19	[−0.41, 0.02]	−0.12	[−0.26, 0.02]	.01	[−0.02, 0.04]	−0.06	
Company size	0.27**	[0.11, 0.42]	0.23	[0.10, 0.37]	.05	[−0.01, 0.11]	.22**	
Experience with cyber risks	−0.05	[−0.38, 0.27]	−0.02	[−0.16, 0.11]	.00	[−0.01, 0.01]	−0.04	
								$R^2 = 0.060^*$ 95% CI[.00,.11]
(Intercept)	1.55*	[0.05, 3.05]						
Age	−0.01	[−0.02, 0.01]	−0.06	[−0.19, 0.06]	.00	[−0.01, 0.02]	.04	
Gender	0.10	[−0.11, 0.31]	0.06	[−0.07, 0.19]	.00	[−0.01, 0.02]	.02	
Education	−0.18	[−0.39, 0.02]	−0.11	[−0.24, 0.01]	.01	[−0.01, 0.04]	−0.06	
Company size	0.25**	[0.11, 0.39]	0.22	[0.10, 0.34]	.04	[−0.00, 0.09]	.22**	
Experience with cyber risks	−0.06	[−0.37, 0.24]	−0.03	[−0.15, 0.10]	.00	[−0.01, 0.01]	−0.04	
Extraversion	−0.08	[−0.21, 0.04]	−0.10	[−0.24, 0.05]	.01	[−0.01, 0.02]	.10	
Agreeableness	0.24**	[0.08, 0.40]	0.20	[0.07, 0.33]	.03	[−0.01, 0.07]	.31**	
Conscientiousness	0.35**	[0.21, 0.49]	0.35	[0.21, 0.50]	.08	[.02, 0.15]	.36**	
Openness	0.14*	[0.01, 0.27]	0.14	[0.01, 0.27]	.02	[−0.01, 0.05]	.23**	
Neuroticism	0.05	[−0.08, 0.19]	0.06	[−0.10, 0.22]	.00	[−0.01, 0.01]	−0.13*	
								$R^2 = 0.257^{**}$ 95% CI[.13,.32]
(Intercept)	1.28	[−0.19, 2.75]						
Age	−0.01	[−0.02, 0.00]						
Gender	0.11	[−0.09, 0.31]						
Education	−0.10	[−0.30, 0.09]						
Company size	0.25**	[0.11, 0.38]						
Experience with cyber risks	−0.09	[−0.38, 0.19]						
Extraversion	−0.07	[−0.19, 0.05]						
Agreeableness	0.24**	[0.08, 0.39]						
Conscientiousness	0.30**	[0.16, 0.44]						
Openness	0.10	[−0.03, 0.23]						
Neuroticism	0.06	[−0.07, 0.19]						
Initial internalization profile	0.64**	[0.29, 0.98]						
Autonomously regulated profile	0.87**	[0.51, 1.23]						
								$R^2 = 0.333^{**}$ 95% CI[.19,.39]

Note: * indicates $p < .05$; ** indicates $p < .01$; Profiles are dummy-coded with amotivated profile as the reference category.

robust predictors. These findings align with previous studies on cybersecurity behavior, which consistently report conscientiousness and, to a lesser extent, agreeableness as the most reliable personality correlates of security-related outcomes, while effects for other Big Five traits tend to be weak, inconsistent, or context-dependent (Gratian et al., 2018; McCormac et al., 2017; Pattinson et al., 2015). The absence of effects for extraversion, neuroticism, and openness in the final regression model reflects the norm-bound and rule-oriented nature of ISSP compliance, which places limited functional demands on traits related to social expressiveness, affective reactivity, or exploration. However, beyond these dispositional influences, *motivational profiles* accounted for additional variance. More specifically, the *autonomously regulated profile* showed the strongest association with ISSP compliance intention, exceeding the effect of the *initial internalized profile*. This finding aligns with prior research suggesting that employees' compliance is more strongly driven by intrinsic and value-based motivations than by deterrence-related considerations (Son, 2011). From an SDT perspective, this is theoretically coherent, as dominant identified and intrinsic regulation imply a self-endorsed and value-congruent engagement in cybersecure behavior, which is known to foster stable and enduring intentions. Importantly, ISSP compliance intention thus appears to depend less on stable personality traits alone but additionally on the quality of domain-specific motivational regulation. This extends personality-based accounts with a theoretically grounded motivational perspective.

4.2. Limitations

The present study provides valuable insights into the motivational and dispositional aspects of ISSP compliance intention. However,

several limitations should be considered when interpreting its findings.

One limitation is our exclusive reliance on self-reported data, which introduces the *risk of social desirability bias* (Hewitt and White, 2022). Although previous studies have documented a systematic link between intention and behavior (Pahnila et al., 2007), it is still possible that participants overreported their willingness to comply with ISSP in order to appear more diligent or capable, even under conditions of anonymity. Consequently, future studies should develop context-specific items that capture various security threat scenarios and incorporate action-oriented assessment methods, such as situational judgment tests, to examine how employees respond to security-relevant challenges in realistic contexts (Keller and Warwas, 2026).

Another limitation concerns the *content validity* and *cross-lingual equivalence* of the questionnaire. Although all scales were previously validated in isolation, we did not conduct a formal content-validity assessment for the combined instrument. Moreover, while the questionnaire was translated into German by bilingual experts, no formal back-translation was conducted to verify the conceptual equivalence of the BFI-44 and the ISSP compliance intention scale. Future research should therefore include fine-grained content-validity checks (e.g., expert reviews or mapping procedures) as well as systematic back-translation procedures to ensure adequate construct coverage and conceptual equivalence in specific organizational contexts.

An additional limitation concerns our *recruitment* via Prolific. While this platform enables efficient access to participants, such online panel samples may not be fully representative of employees in German SMEs (e.g., with respect to occupational roles, organizational context, or digital affinity). Therefore, the generalizability of our findings to the broader population of SME employees and to actual workplace behavior should be interpreted with caution.

Moreover, the study's cross-sectional design makes it difficult to establish *causal relationships* between personality characteristics, motivational modes, and ISSP compliance intention. Research designs that allow at least a temporal ordering of variables are needed to draw unambiguous conclusions.

The study deliberately focused on office workers from German SMEs, since this group differs from employees in larger organizations and other professional domains in terms of work routines, cybersecurity resources, and exposure to digital threats. Findings from this clear yet restricted setting should be interpreted against the backdrop of the *cultural and institutional framework* of German SMEs. While the assumptions of both SDT and trait theory are not inherently country-specific, the prevalence of motivational profiles and the magnitude of their associations with ISSP compliance intentions may vary across settings with different rule-adherence norms, risk perceptions, or enforcement climates. Furthermore, the sample included a predominantly young population, which may not capture age-related differences in security motivation or compliance intentions. Older or more experienced employees, for instance, may perceive risks differently or rely on habitualized work activities to a greater extent, leading to potentially distinct motivational patterns (Chua et al., 2018). Future research should therefore examine more diverse samples across sectors, age groups, and organizational structures. Optimally, it would include cross-national replications to examine measurement invariance, compare profile solutions, and contrast predictor–outcome links across cultural and regulatory environments.

Finally, by aiming to elaborate and examine *psychological aspects* of ISSP compliance intention among the staff, the study omitted *organizational determinants*. Top management commitment, security communication, responsibilities, leadership style, and technical support availability (D'Arcy and Greene, 2014) may well affect compliance intention directly (such as through role models) or indirectly (such as through motivational incentives). Incorporating these variables in future studies would provide an even more comprehensive understanding of how psychological and organizational factors jointly influence ISSP compliance intention.

4.3. Practical implications

The findings from the exploratory, integrative approach adopted in the present study suggest that conscientiousness and agreeableness function as relatively stable dispositional factors that may shape employees' baseline propensities for rule adherence and norm-oriented behavior such as ISSP compliance. This carries practical relevance for organizational *risk assessment*. These traits cannot be readily changed through targeted interventions. Therefore, they should be considered as relatively stable dispositional factors associated with compliance-related risk. However, the assessment and use of personality-related data must be handled with caution, as such data raise important privacy, fairness, and ethical concerns and should therefore be collected transparently, used only for legitimate and job-related purposes, and not be used to justify discriminatory personnel decisions (Tursunbayeva et al., 2022). Within these boundaries, lower levels of conscientiousness or agreeableness, if assessed in practice through validated and transparently administered personality self-report instruments, may signal a higher likelihood of poor ISSP compliance and, thus, point to a greater need for *clear structures, explicit expectations, and consistent guidance* (Shropshire et al., 2015).

Whereas personality traits help to explain who is generally more inclined toward compliant behavior, motivation accounts for whether rule-congruent (cybersecure) behavior occurs rather in a self-regulated, internalized mode or remains strictly externally controlled and contingent. Such insights can help to identify further leverage points for organizational intervention.

Due to their internalized action regulation, it can be assumed that autonomously regulated employees willingly and effectively contribute

to protecting their organization against cyber threats. They are also well-placed to support security training by sharing best practices and encouraging colleagues to follow the ISSP in their own day-to-day work. Although the segment of employees with an amotivated profile may appear negligibly small in the present sample, it is highly relevant from a risk-management perspective. The absence of both external and internal motivators increases the likelihood of inadvertent policy violations. Because a single point of failure can compromise an organization's overall security posture, reducing the proportion of amotivated employees becomes a critical organizational task. The SDT offers a compelling framework for explaining how companies can facilitate the internalization of cybersecure behavior on the part of their staff.

Drawing on an *organismic perspective*, SDT posits that individuals have an innate drive to grow, integrate new experiences, and align their actions with meaningful personal goals (Deci and Ryan, 1985). At the same time, the theory's *dialectical perspective* emphasizes that this developmental process unfolds through continuous interaction with the social environment (Deci and Ryan, 1993). In particular, the extent to which the work environment *supports basic psychological needs* for autonomy, competence, and relatedness (Deci and Ryan, 2000) affects the extent to which employees internalize cybersecurity practices. Employees are more likely to adopt secure practices for self-endorsed reasons when they understand the rationale behind security measures, feel capable of implementing them, and experience cybersecurity as a shared professional responsibility (Gangire et al., 2021). These autonomous modes of motivation foster deeper engagement, greater persistence, and more stable behavioral patterns over time (Deci et al., 2017).

Building on this theoretical foundation helps to delineate reasonably effective organizational measures. If secure behavior is to be internalized rather than enforced, interventions must be designed in ways that actively support employees' basic psychological needs.

To support *competence*, training formats should illustrate which practices secure behavior entails exactly, why these practices matter, and how they mitigate concrete risks in daily work. Relatable examples, realistic scenarios, and clear explanations foster a sense of mastery, enabling employees to perceive secure behavior as meaningful, feasible, and directly connected to their professional responsibilities (Kam et al., 2020).

A second lever is *relatedness support*. Employees are more likely to behave securely when they experience cybersecurity as a shared commitment rather than an individual burden. Leaders play a pivotal role here. By modeling secure practices and articulating their ethical and strategic value, leaders can cultivate a climate in which security is perceived as a collective endeavor (Xue et al., 2021).

Finally, practices that support *autonomy* can increase employees' willingness to act securely by fostering a sense of ownership over security responsibilities. When employees have meaningful opportunities to influence their security environment, they are more likely to view security requirements as personally endorsed rather than externally imposed (Gerdenitsch et al., 2023). These opportunities include providing feedback on the practicality of existing security rules, identifying security-relevant vulnerabilities in everyday work routines, and collaborating on developing feasible and effective security procedures. These participatory structures cultivate personal responsibility, which fosters autonomous motivation and has been shown to strengthen compliance with ISSP directly. At the same time, these structures reduce the demotivating effects of overly punitive or controlling approaches, which can undermine self-determination and weaken long-term adherence to secure behaviors (Hong and Xu, 2021). Together, these implications highlight that fostering ISSP compliance requires more than designing SETA measures or sanctions for noncompliance. Rather, our findings encourage cultivating organizational environments that support employees' psychological needs, thus allowing secure behavior to become personally meaningful, socially embedded, and sustainably enacted.

5. Conclusion

Given the ongoing digitalization of work environments and the increasing demands placed on employees to safeguard sensitive data, this study aimed to clarify the relative contribution of personality traits and domain-specific motivation to ISSP compliance intentions among office-based employees in SMEs. The findings underscore the need to align training and communication strategies with employees' regulation modes of cybersecurity behaviors, as qualitatively distinct motivational profiles emerged and explained incremental variance in ISSP compliance intention beyond structural and personality characteristics. Theoretically, these results extend the previously established trait-based accounts of (non-)compliance by integrating an SDT perspective on motivational quality and internalization into cybersecurity research. Practically, tailoring interventions to these underlying drivers allows organizations to design measures that more effectively promote responsible action against cyber risks. In particular, the study points to design features of workplaces and training that facilitate internalization processes among staff. This means fostering long-term voluntary compliance by emphasizing the internalization of security policies instead of relying solely on external enforcement mechanisms. This approach can strengthen employees' autonomous commitment to secure behavior, supporting the development of a robust, proactive security

culture.

CRedit authorship contribution statement

Thomas Keller: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Julia Warwas:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology. **Josef Guggemos:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Verena Zengerle:** Writing – review & editing, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare, that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Publishing fees are supported by the funding program Open Access Publishing of the University of Hohenheim.

Appendix A. Results of the confirmatory factor analysis

Table A1

Fit Indices of competing CFA models for the MWMS.

Model	CFI	TLI	RMSEA	SRMR
Model 1: Single-Factor Model	0.481	0.416	0.196	0.144
Model 2: Three-Factor Model	0.706	0.663	0.149	0.132
Model 3: Five-Factor Model (original)	0.872	0.845	0.101	0.08
Model 4: Five-Factor Model (differentiated external regulation)	0.910	0.891	0.085	0.069
Model 5: Five-Factor Model (differentiated external and introjected regulation)	0.947	0.934	0.065	0.064

Note. CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Square Residual.

Model 1: unidimensional, with all items loading onto a single overarching factor. Fit indices suggest that an undifferentiated conceptualization of motivational modes is not empirically tenable (Table A1).

Model 2: three-factor model, distinguishing *amotivation* from controlled motivational modes (*external* and *introjected*) and autonomous motivational modes (*identified* and *intrinsic*). This theoretically grounded structure improved the model fit but still failed acceptable levels.

Model 3: a five-factor structure in which each mode of action regulation – *amotivation*, *external regulation*, *introjected regulation*, *identified regulation*, and *intrinsic regulation* – was modeled as a distinct first-order factor. Fully consistent with SDT. Fit indices improved, though there was still room for optimization.

Model 4: an even more nuanced structure in which external regulation was split into two subdimensions: *material* (e.g., rewards and punishments) and *social* (e.g., social pressure and expectations). Subdimensions were modeled as separate first-order factors, both loading onto the higher-order factor external regulation. This hierarchical specification provided an even more accurate representation of external motivational drivers, resulting in further improvements of model fit.

Model 5: an extended hierarchical model further differentiating introjected regulation into two subdimensions: *approach-oriented* and *avoidance-oriented* introjection. These subdimensions were specified as separate first-order factors loading onto a second-order factor of introjected regulation. Analogous to the hierarchical structure of external regulation, this specification captures the theoretically meaningful heterogeneity within introjected motivation and resulted in additional improvements in model fit.

Appendix B. Results of the split-half cross-validation

Table B1

Split-half LPA model fit indices (Subsample 1).

No. of Profiles	AIC	BIC	Adjusted BIC	Entropy	Min. profile n	Min. probability	p(BLRT)
1	1563	1590	1559	1.000	1.000	1.000	—
2	1399	1442	1391	0.933	0.156	0.936	< 0.01
3	1303	1362	1293	0.875	0.119	0.934	< 0.01
4	1281	1357	1268	0.863	0.101	0.874	< 0.01
5	1272	1363	1256	0.835	0.101	0.830	< 0.01

Split-half LPA model fit indices (Subsample 2).

No. of Profiles	AIC	BIC	Adjusted BIC	Entropy	Min. profile n	Min. probability	p(BLRT)
1	1436	1463	1431	1.000	1.000	1.000	—
2	1288	1331	1281	0.843	0.486	0.940	< 0.01
3	1256	1315	1246	0.830	0.257	0.885	< 0.01
4	1245	1321	1232	0.809	0.101	0.849	< 0.01
5	1230	1321	1214	0.881	0.046	0.846	< 0.01

Note. The sample was randomly divided into two subsamples, which were then analyzed independently. The preferred number of profiles differed slightly across subsamples (three versus four profiles), indicating that class enumeration was somewhat sensitive to sampling variability.

Table B2

Aligned profile means and split-half differences for the three-profile solution.

Profile	Amotivation	External	Introjected	Identified	Intrinsic	n	Δ Amotivation	Δ External	Δ Introjected	Δ Identified	Δ Intrinsic
1	−0.63	0.56	0.76	0.80	0.69	38	−0.10	0.06	−0.18	−0.22	−0.20
2	0.19	−0.21	−0.34	−0.33	−0.28	58	0.08	−0.15	−0.41	−0.38	−0.12
3	1.51	−1.30	−1.95	−2.15	−1.56	13	1.16	−0.97	−1.11	−1.20	−0.87

Aligned profile means and split-half differences for the four-profile solution.

Profile	Amotivation	External	Introjected	Identified	Intrinsic	n	Δ Amotivation	Δ External	Δ Introjected	Δ Identified	Δ Intrinsic
1	−0.87	0.92	1.23	1.29	1.45	11	−0.26	0.22	0.04	0.10	0.57
2	−0.40	0.20	0.37	0.44	0.12	45	−0.31	−0.04	0.03	−0.03	−0.39
3	0.44	−0.28	−0.56	−0.59	−0.30	42	0.39	0.03	−0.26	−0.19	0.23
4	1.46	−1.39	−2.08	−2.32	−1.67	11	0.49	−0.82	−0.81	−1.05	−0.83

Note. A detailed comparison of aligned profile means showed overall greater stability in profile shape for the three-profile solution across subsamples. In contrast, the four-profile solution suggested an unstable split of an extreme profile into smaller subgroups, reflected in larger cross-split deviations in profile means, particularly for the smallest profiles.

Appendix C. Cybersecurity multidimensional work motivation scale for cybersecurity behavior

Instructions

There are different reasons why you behave in an information-secure manner when performing your daily tasks in your organization – or why you do not (Table C1). This is because activities aimed at protecting sensitive data from unauthorized access, loss, theft, or damage can often be time-consuming and challenging. Please indicate to what extent you agree with each of the following statements as reasons for why you behave in an information-secure way at work.

Table C1

Items of the cybersecurity multidimensional work motivation scale for cybersecurity behavior.

Construct	Item Code	Item
Amotivation	AMO_1	I don't make an effort because I feel like I'm wasting my time with this work.
	AMO_2	I don't make an effort because this work isn't worth the effort.
	AMO_3	I don't make an effort because I don't know why I'm doing this work, it's pointless.
Extrinsic regulation	EXT_1	I do make an effort because my employer or supervisors only reward me financially if I do.
	EXT_2	I do make an effort because my employer or supervisors offer me greater job security.
	EXT_3	I do make an effort because otherwise I risk losing my job.
	EXT_4	I do make an effort to receive recognition from my superiors and colleagues.
	EXT_5	I do make an effort to gain more respect from my supervisors and colleagues.
	EXT_6	I do make an effort to avoid being criticized by my superiors and colleagues.
Introjected regulation	INT_1	I do make an effort because I have to prove to myself that I'm capable of it.

(continued on next page)

Table C1 (continued)

Construct	Item Code	Item
Identified regulation	INT_2	I do make an effort because it makes me feel proud of myself.
	INT_3	I do make an effort because otherwise I will feel ashamed of myself.
	INT_4	I do make an effort because otherwise I will feel bad about myself.
	IDN_1	I do make an effort because it is important to me personally.
	IDN_2	I do make an effort because it aligns with my values.
Intrinsic motivation	IDN_3	I do make an effort because it means a lot to me personally.
	INTRI_1	I do make an effort because I enjoy it.
	INTRI_2	I do make an effort because I find it exciting.
	INTRI_3	I do make an effort because I find it interesting.

Note. Participants indicated their agreement on a 6-point Likert scale (1 = "does not apply at all", 6 = "applies completely").

Data availability

The dataset used in this study is available at OSF:
[Motivational Qualities \(Original data\)](#) (Open Science Framework)

References

- Adriko, R., Nurse, J.R.C., 2026. Cybersecurity and cyber insurance for small to medium-sized enterprises (SMEs): perceptions, challenges and decision-making dynamics. *Comput. Secur.* 163, 104818. <https://doi.org/10.1016/j.cose.2025.104818>.
- Alotaibi, M., Furnell, S., Clarke, N., 2016. Information security policies: a review of challenges and influencing factors. In: *Proceedings of the 2016 11th International Conference for Internet Technology and Secured Transactions (ICITST)*. IEEE, pp. 352–358. <https://doi.org/10.1109/ICITST.2016.7856729>.
- Balogopal, N., Mathew, S.K., 2024. Exploring the factors influencing information security policy compliance and violations: a systematic literature review. *Comput. Secur.* 147, 104062. <https://doi.org/10.1016/j.cose.2024.104062>.
- Bauer, J., 2022. A primer to latent profile and latent class analysis. In: Goller, M., Kyndt, E., Paloniemi, S., Damşa, C. (Eds.), *Methods For Researching Professional Learning and Development: Challenges, Applications and Empirical Illustrations* (pp. 243–268). Springer International Publishing. https://doi.org/10.1007/978-3-031-08518-5_11.
- Beaton, D.E., Bombardier, C., Guillemin, F., Ferraz, M.B., 2000. Guidelines for the process of cross-cultural adaptation of self-report measures. *Spine* 25 (24), 3186–3191. <https://doi.org/10.1097/00007632-200012150-00014>.
- Bjerg Hall-Andersen, L., Broberg, O., 2014. Learning processes across knowledge domains. *J. Workplace Learn.* 26 (2), 91–108. <https://doi.org/10.1108/JWL-08-2013-0053>.
- Boss, S.R., Galletta, D.F., Lowry, P.B., Moody, G.D., Polak, P., 2015. What do systems users have to fear? Using fear appeals to engender threats and fear that motivate protective behaviors in users. *MIS Q.* 39 (4), 865–885. <https://doi.org/10.25300/MISQ/2015/39.4.6>.
- Bouraffa, T., Hui, K.L., 2025. Regulating information and network security: review and challenges. *ACM Comput. Surv.* 57 (5), 1–38. <https://doi.org/10.1145/3711124>.
- Broderick, M., Schwartz, A., McDonald, K., 2025. A scoping review of the ways gatekeepers may hinder or promote opportunities for people with intellectual and/or developmental disabilities to learn about research participation. *J. Intellect. Disabil. Res.* 69 (5), 329–344. <https://doi.org/10.1111/jir.13216>.
- Butavicius, M., Taib, R., Han, S.J., 2022. Why people keep falling for phishing scams: effects of time pressure and deception cues on detection of phishing emails. *Comput. Secur.* 123, 102937. <https://doi.org/10.1016/j.cose.2022.102937>.
- Camillo, M., 2017. Cybersecurity: risks and management of risks for global banks and financial institutions. *J. Risk Manag. Financ. Inst.* 10 (2), 196. <https://doi.org/10.69554/EPYV4777>.
- Chen, Y., Xia, W., Cousins, K., 2022. Voluntary and instrumental information security policy compliance. An integrated view of prosocial motivation, self regulation and deterrence. *Comput. Secur.* 113, 102568. <https://doi.org/10.1016/j.cose.2021.102568>.
- Chua, H.N., Wong, S.F., Low, Y.C., Chang, Y., 2018. Impact of employees' demographic characteristics on the awareness and compliance of information security policy in organizations. *Telemat. Inform.* 35 (6), 1770–1780. <https://doi.org/10.1016/j.tele.2018.05.005>.
- Conway, J.M., Lance, C.E., 2010. What reviewers should expect from authors regarding common method bias in organizational research. *J. Bus. Psychol.* 25 (3), 325–334. <https://doi.org/10.1007/s10869-010-9181-6>.
- Cram, W.A., Proudfoot, J.G., D'Arcy, J., 2017. Organizational information security policies: a review and research framework. *Eur. J. Inf. Syst.* 26 (6), 605–641. <https://doi.org/10.1057/s41303-017-0059-9>.
- D'Arcy, J., Greene, G., 2014. Security culture and the employment relationship as drivers of employees' security compliance. *Inf. Comput. Secur.* 22 (5), 474–489. <https://doi.org/10.1108/IMCS-08-2013-0057>.
- D'Arcy, J., Hovav, A., Galletta, D., 2009. User awareness of security countermeasures and its impact on information systems misuse: a deterrence approach. *Inf. Syst. Res.* 20 (1), 79–98. <https://doi.org/10.1287/isre.1070.0160>.
- Deci, E.L., Olafsen, A.H., Ryan, R.M., 2017. Self-determination theory in work organizations: the state of a science. *Annu. Rev. Organ. Psychol. Behav.* 4, 19–43. <https://doi.org/10.1146/annurev-orgpsych-032516-113108>.
- Deci, E.L., Ryan, R.M., 1985. *Intrinsic Motivation and Self-Determination in Human Behavior*. Plenum, New York.
- Deci, E.L., Ryan, R.M., 1993. Die selbstbestimmungstheorie der motivation und ihre bedeutung für die pädagogik. *Z. Pädagogik* 39 (2), 223–238.
- Deci, E.L., Ryan, R.M., 2000. The "what" and "why" of goal pursuits: human needs and the self-determination of behavior. *Psychol. Inq.* 11 (4), 227–268.
- Dhillon, G., Backhouse, J., 2001. Current directions in IS security research: towards socio-organizational perspectives. *Inf. Syst. J.* 11 (2), 127–153.
- Diamant, I., Rusou, Z., 2021. Measuring failures proneness: scale development and preliminary validations. *Front. Psychol.* 12, 757051. <https://doi.org/10.3389/fpsyg.2021.757051>.
- Diener, E., Lucas, R.E., 2026. Personality traits. In: Biswas-Diener, R., Diener, E. (Eds.), *Noba Textbook Series: Psychology*. DEF Publishers, Champaign, IL. Retrieved from. <http://noba.to/96u8ecgw>.
- Dunn, A.M., Heggstad, E.D., Shanock, L.R., Theilgard, N., 2018. Intra-individual response variability as an indicator of insufficient effort responding: comparison to other indicators and relationships with individual differences. *J. Bus. Psychol.* 33 (1), 105–121. <https://doi.org/10.1007/s10869-016-9479-0>.
- ENISA. (2021). ENISA threat landscape 2021. April 2020 to Mid-July 2021 (Issue October). [10.2824/324797](https://doi.org/10.2824/324797).
- Fetchnhauer, D., 2011. *Persönlichkeitspsychologie: Ein Lehrbuch*. Springer-Verlag, Berlin, Heidelberg.
- Frank, M., Kohn, V., 2023. Understanding extra-role security behaviors: an integration of self-determination theory and construal level theory. *Comput. Secur.* 132, 103386. <https://doi.org/10.1016/j.cose.2023.103386>.
- Furnell, S., Rajendran, A., 2012. Understanding the influences on information security behaviour. *Comput. Fraud Secur.* 2012 (3), 12–15. [https://doi.org/10.1016/S1361-3723\(12\)70053-2](https://doi.org/10.1016/S1361-3723(12)70053-2).
- Gagné, M., Forest, J., Gilbert, M.H., Aubé, C., Morin, E., Malorni, A., 2015. The multidimensional work motivation scale: validation evidence in seven languages and nine countries. *Eur. J. Work Organ. Psychol.* 24 (2), 178–196. <https://doi.org/10.1080/1359432X.2013.877892>.
- Gallo, L., Gentile, D., Ruggiero, S., Botta, A., Ventre, G., 2024. The human factor in phishing: collecting and analyzing user behavior when reading emails. *Comput. Secur.* 139, 103671. <https://doi.org/10.1016/j.cose.2023.103671>.
- Gangire, Y., Da Veiga, A., Herselman, M., 2021. Assessing information security behaviour: a self-determination theory perspective. *Inf. Comput. Secur.* 29 (4), 625–646. <https://doi.org/10.1108/ICS-11-2020-0179>.
- Gerdenitsch, C., Wurhofer, D., Tscheligi, M., 2023. Working conditions and cybersecurity: time pressure, autonomy and threat appraisal shaping employees' security behavior. *Cyberpsychol. J. Psychosoc. Res. Cyberspace* 17 (4), 7. <https://doi.org/10.5817/CP2023-4-7>.
- Gillet, N., Morin, A.J., Ndiaye, A., Colombat, P., Fouquereau, E., 2020. A test of work motivation profile similarity across four distinct samples of employees. *J. Occup. Organ. Psychol.* 93 (4), 988–1030. <https://doi.org/10.1111/joop.12322>.
- Grandhi, S.R., Still, J.D., 2025. The Big Five in action: a systematic review of personality, cyber awareness, and behaviors. In: Moallem, A. (Ed.), *Proceedings of the HCI for Cybersecurity, Privacy and Trust: 7th International Conference, HCI-CPT 2025, Held as Part of the 27th HCI International Conference, HCII 2025, Gothenburg, Sweden. Springer*, pp. 17–34. https://doi.org/10.1007/978-3-031-92833-8_2. June 22–27, 2025Proceedings, Part I.
- Gratani, M., Bandi, S., Cukier, M., Dykstra, J., Ginther, A., 2018. Correlating human traits and cyber security behavior intentions. *Comput. Secur.* 73, 345–358. <https://doi.org/10.1016/j.cose.2017.11.015>.
- Hair, J.F., Black, W.C., Babin, B.J., Anderson, R.E., 2019. *Multivariate Data Analysis*, 8th ed. Cengage.
- Hair, J.F., Ringle, C.M., Sarstedt, M., 2011. PLS-SEM: indeed a silver bullet. *J. Mark. Theory Pract.* 19 (2), 139–152. <https://doi.org/10.2753/mtp1069-6679190202>.
- He, Y., Lei, P., 2025. Differential pathways from personality to risk-taking: how extraversion and negative emotionality shape decision-making through overconfidence. *Front. Psychol.* 16, 1537658. <https://doi.org/10.3389/fpsyg.2025.1537658>.
- Heckhausen, J., Heckhausen, H., 2018. Motivation and action: introduction and overview. In: Heckhausen, J., Heckhausen, H. (Eds.), *Motivation and Action*. Springer, pp. 1–14. https://doi.org/10.1007/978-3-319-65094-4_1.

- Herath, T., Rao, H.R., 2009. Protection motivation and deterrence: a framework for security policy compliance in organisations. *Eur. J. Inf. Syst.* 18 (2), 106–125. <https://doi.org/10.1057/ejis.2009.6>.
- Hewitt, B., White, G.L., 2022. Optimistic bias and exposure affect security incidents on home computer. *J. Comput. Inf. Syst.* 62 (1), 50–60. <https://doi.org/10.1080/08874417.2020.1728737>.
- Hong, Y., Xu, M., 2021. Autonomous motivation and information security policy compliance: role of job satisfaction, responsibility, and deterrence. *J. Organ. End User Comput.* 33 (6), 1–17. <https://doi.org/10.4018/JOEUC.20211101.0a9>.
- Howard, J.L., Gagné, M., Morin, A.J.S., Van den Broeck, A., 2016. Motivation profiles at work: a self-determination theory approach. *J. Vocat. Behav.* 95–96, 74–89. <https://doi.org/10.1016/j.jvb.2016.07.004>.
- Hu, L.T., Bentler, P.M., 1999. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct. Equ. Model.* 6 (1), 1–55. <https://doi.org/10.1080/10705519909540118>.
- Humayun, M., Niazi, M., Jhanjhi, N.Z., Alshayeb, M., Mahmood, S., 2020. Cyber security threats and vulnerabilities: a systematic mapping study. *Arab. J. Sci. Eng.* 45 (4), 3171–3189. <https://doi.org/10.1007/s13369-019-04319-2>.
- IBM Security, 2024. Cost of a Data Breach Report 2024. IBM.
- Ifinedo, P., 2014. Information systems security policy compliance: an empirical study of the effects of socialisation, influence, and cognition. *Inf. Manag.* 51 (1), 69–79. <https://doi.org/10.1016/j.im.2013.10.001>.
- Jach, H.K., Smillie, L.D., 2020. Testing the information-seeking theory of openness/intellect. *Eur. J. Personal.* 35 (1), 103–119. <https://doi.org/10.1002/per.2271>.
- Jani, D., Han, H., 2013. Personality, social comparison, consumption emotions, satisfaction, and behavioral intentions: how do these and other factors relate in a hotel setting? *Int. J. Contemp. Hosp. Manag.* 25 (7), 970–993.
- Johannsen, A., Kant, D., Creutzburg, R., 2020. Measuring IT security, compliance and data governance within small and medium-sized IT enterprises. *Electron. Imaging* 32 (3). <https://doi.org/10.2352/ISSN.2470-1173.2020.3.MOBMU-252>, 252–1–252–11.
- John, O.P., Donahue, E.M., Kentle, R.L., 1991. The "Big Five" Inventory—Versions 4a and 54. University of California, Berkeley, Institute of Personality and Social Research, Berkeley.
- John, O.P., Srivastava, S., 1999. The Big-five trait taxonomy: history, measurement, and theoretical perspectives. In: Pervin, L.A., John, O.P. (Eds.), *Handbook of Personality: Theory and Research*, 2nd ed. Guilford Press, New York, pp. 102–138.
- Johnston, A.C., Warkentin, M., 2010. Fear appeals and information security behaviors: an empirical study. *MIS Q.* 34 (3), 549–566. <https://doi.org/10.2307/25750691>.
- Jouini, M., Rabai, L.B.A., Aissa, A.B., 2014. Classification of security threats in information systems. *Procedia Comput. Sci.* 32, 489–496. <https://doi.org/10.1016/j.procs.2014.05.452>.
- Jung, T., Wickrama, K.A.S., 2008. An introduction to latent class growth analysis and growth mixture modeling. *Soc. Personal. Psychol. Compass* 2 (1), 302–317. <https://doi.org/10.1111/j.1751-9004.2007.00054.x>.
- Kam, H.J., Menard, P., Ormond, D., Crossler, R.E., 2020. Cultivating cybersecurity learning: an integration of self-determination and flow. *Comput. Secur.* 96, 101875. <https://doi.org/10.1016/j.cose.2020.101875>.
- Keller, T., Warwas, J., 2026. Entwicklung und Validierung eines Situational Judgment Tests zur Erfassung der Informationssicherheitskompetenzen von Berufseinsteiger_innen an typischen Büroarbeitsplätzen. *Diagnostica* 72 (2), 92–104. <https://doi.org/10.1026/0012-1924/a000368>.
- Kirlappos, I., Parkin, S., Sasse, A., 2014. Learning from “shadow security”: why understanding non-compliant behaviours provides the basis for effective security. In: *Proceedings of the 2014 Workshop on Usable Security (USEC '14)*. Internet Society, pp. 1–10.
- Kranz, J.J., Haeussinger, F.J., 2014. Why deterrence is not enough: the role of endogenous motivations on employees' information security behavior. In: Myers, M. D., Straub, D.W. (Eds.), *Proceedings of the International Conference on Information Systems - Building a Better World through Information Systems*. Auckland, New Zealand. Association for Information Systems, pp. 1–14 (CIS) 2014.
- López-Aguilar, P., Urruela, C., Batista, E., Machin, J., Solanas, A., 2025. Phishing vulnerability and personality traits: insights from a systematic review. *Comput. Hum. Behav. Rep.* 20, 100784. <https://doi.org/10.1016/j.chbr.2025.100784>.
- Lubke, G.H., Muthén, B., 2005. Investigating population heterogeneity with factor mixture models. *Psychol. Methods* 10 (1), 21–39. <https://doi.org/10.1037/1082-989X.10.1.21>.
- Masyn, K.E., 2013. Latent class analysis and finite mixture modeling. In: Little, T.D. (Ed.), *The Oxford Handbook of Quantitative Methods*, The Oxford Handbook of Quantitative Methods, 2. Oxford University Press, pp. 551–611.
- McCormac, A., Zwaans, T., Parsons, K., Calic, D., Butavicius, M., Pattinson, M., 2017. Individual differences and information security awareness. *Comput. Hum. Behav.* 69, 151–156. <https://doi.org/10.1016/j.chb.2016.11.065>.
- McCrae, R.R., John, O.P., 1992. An introduction to the five-factor model and its applications. *J. Personal.* 60 (2), 175–215. <https://doi.org/10.1111/j.1467-6494.1992.tb00970.x>.
- Meade, A.W., Craig, S.B., 2012. Identifying careless responses in survey data. *Psychol. Methods* 17 (3), 437–455. <https://doi.org/10.1037/a0028085>.
- Michel, Y.A., Steinmayr, R., Frenzel, A.C., Ziegler, M., 2022. Unpacking domain-specific achievement motivation: the role of contextualising items for test–criterion correlations. *Educ. Psychol.* 42 (4), 501–525. <https://doi.org/10.1080/01443410.2020.1713303>.
- Miller, B.K., Simmering, M.J., 2023. Attitude toward the color blue: an ideal marker variable. *Organ. Res. Methods* 26 (3), 409–440. <https://doi.org/10.1177/10944281221075361>.
- Morin, A.J.S., McLarnon, M.J.W., Litalien, D., 2020. Mixture modeling for organizational behavior research. In: Griep, Y., Hansen, S.D. (Eds.), *Handbook on the Temporal Dynamics of Organizational Behavior*. Edward Elgar Publishing, pp. 351–379. <https://doi.org/10.4337/9781788974387.00031>.
- Muthén, B., 2004. Latent variable analysis. In: *The Sage Handbook of Quantitative Methodology for the Social Sciences*, 345. SAGE Publications, Inc, pp. 106–109. <https://doi.org/10.4135/9781412986311.n19>.
- Neyer, F.J., Asendorpf, J.B., 2024. *Psychologie Der Persönlichkeit*. Springer, Berlin, Heidelberg. <https://doi.org/10.1007/978-3-662-67385-0> (7. vollst. überarb. Aufl.).
- Nunnally, J.C., Bernstein, I.H., 1994. *Psychometric Theory*, 3rd ed. McGraw-Hill, New York.
- Nylund-Gibson, K., Choi, A.Y., 2018. Ten frequently asked questions about latent class analysis. *Trans. Issues Psychol. Sci.* 4 (4), 440–461. <https://doi.org/10.1037/tps0000176>.
- Ozer, D.J., Benet-Martínez, V., 2006. Personality and the prediction of consequential outcomes. *Annu. Rev. Psychol.* 57, 401–421. <https://doi.org/10.1146/annurev.psych.57.102904.190127>.
- Padayachee, K., 2012. Taxonomy of compliant information security behavior. *Comput. Secur.* 31 (5), 673–680. <https://doi.org/10.1016/j.cose.2012.04.004>.
- Pahnila, S., Siponen, M., Mahmood, A., 2007. Employees' behavior towards IS security policy compliance. In: *Proceedings of the 40th Annual Hawaii International Conference on System Sciences (HICSS)*. USA, Hawaii. <https://doi.org/10.1109/HICSS.2007.206>.
- Pattinson, M., Butavicius, M., Parsons, K., McCormac, A., Calic, D., 2015. Factors that influence information security behavior: an Australian web-based study. *Hum. Asp. Inf. Secur. Priv. Trust* 231–241. https://doi.org/10.1007/978-3-319-20376-8_21.
- Peer, E., Rothschild, D., Gordon, A., Evernden, Z., Damer, E., 2022. Data quality of platforms and panels for online behavioral research. *Behav. Res. Methods* 54 (4), 1643–1662. <https://doi.org/10.3758/s13428-021-01694-3>.
- Pinder, C., 2008. *Work Motivation in Organizational Behavior*, 2nd ed. Psychology Press.
- Podsakoff, P.M., MacKenzie, S.B., Lee, J.Y., Podsakoff, N.P., 2003. Common method biases in behavioral research: a critical review of the literature and recommended remedies. *J. Appl. Psychol.* 88 (5), 879–903. <https://doi.org/10.1037/0021-9010.88.5.879>.
- Profeti, S., Toth, F., 2025. Leading targets to comply. Uncertainty issues in the design of “intrinsic motivation-driven policies. *Policy Des. Pract.* 8, 323–337. <https://doi.org/10.1080/25741292.2025.2466297>.
- Rahman, M.R., Basak, S.K., Mahdavi Hezaveh, R., Williams, L., 2025. SoK: an empirical investigation of malware techniques in advanced persistent threat attacks. *Comput. Secur.* 157, 104618. <https://doi.org/10.1016/j.cose.2025.104618>.
- Roccas, S., Sagiv, L., Schwartz, S.H., Knafo, A., 2002. The big five personality factors and personal values. *Personal. Soc. Psychol. Bull.* 28 (6), 789–801. <https://doi.org/10.1177/0146167202289008>.
- Rogers, R.W., 1975. A protection motivation theory of fear appeals and attitude change. *J. Psychol.* 91 (1), 93–114.
- Romaldo Junior, C., Becker, I., Johnson, S., 2023. Unaware, unfunded and uneducated: a systematic review of SME cybersecurity. *ArXiv*. <https://doi.org/10.48550/arXiv.2309.17186>.
- Rosenberg, J.M., Beymer, P.N., Anderson, D.J., van Lissa, C.J., Schmidt, J.A., 2018. tidyLPA: an R package to easily carry out latent profile analysis (LPA) using open-source or commercial software. *J. Open Source Softw.* 3 (30), 978. <https://doi.org/10.21105/joss.00978>.
- Rosseel, Y., 2012. lavaan: an R package for structural equation modeling. *J. Stat. Softw.* 48 (2), 1–36. <https://doi.org/10.18637/jss.v048.i02>.
- Ryan, R.M., Deci, E.L., 2017. Self-Determination Theory: Basic Psychological Needs in Motivation, Development, and Wellness. The Guilford Press. <https://doi.org/10.1521/978.14625/28806>.
- Ryan, R.M., Deci, E.L., 2020. Intrinsic and extrinsic motivation from a self-determination theory perspective: definitions, theory, practices, and future directions. *Contemp. Educ. Psychol.* 61, 101860. <https://doi.org/10.1016/j.cedpsych.2020.101860>.
- Saban, K.A., Rau, S., Wood, C.A., 2021. SME executives' perceptions and the information security preparedness model. *Inf. Comput. Secur.* 29 (2), 263–282. <https://doi.org/10.1108/ICS-01-2020-0014>.
- Scheffer, D., Heckhausen, H., 2018. Trait theories of motivation. In: Heckhausen, J., Heckhausen, H. (Eds.), *Motivation and Action*. Springer, Berlin, Heidelberg, pp. 67–112. https://doi.org/10.1007/978-3-319-65094-4_3.
- Schmeisser, Y., Renström, E.A., Bäck, H., 2021. Who follows the rules during a crisis? Personality traits and trust as predictors of compliance with containment recommendations during the COVID-19 pandemic. *Front. Political Sci.* 3, 739616. <https://doi.org/10.3389/fpos.2021.739616>.
- Shappie, A.T., Dawson, C.A., Debb, S.M., 2020. Personality as a predictor of cybersecurity behavior. *Psychol. Pop. Media* 9 (4), 475–480. <https://doi.org/10.1037/ppm0000247>.
- Shropshire, J., Warkentin, M., Sharma, S., 2015. Personality, attitudes, and intentions: predicting initial adoption of information security behavior. *Comput. Secur.* 49, 177–191. <https://doi.org/10.1016/j.cose.2015.01.002>.
- Siponen, M., Willison, R., 2009. Information security management standards: problems and solutions. *Inf. Manag.* 46, 267–270. <https://doi.org/10.1016/j.im.2008.12.007>.
- Siponen, M., Vance, A., 2010. Neutralization: new insights into the problem of employee information systems security policy violations. *MIS Q.* 34 (3), 487–502. <https://doi.org/10.2307/25750688>.
- Son, J.Y., 2011. Out of fear or desire? Toward a better understanding of employees' motivation to follow IS security policies. *Inf. Manag.* 48 (7), 296–302. <https://doi.org/10.1016/j.im.2011.07.002>.
- Talib, Y.Y.A. (2015). Intrinsic motivation and information systems security policy compliance in organizations [Doctoral Dissertation, Virginia Commonwealth University]. VCU Scholars Compass.

- Tóth-Király, I., Morin, A.J., Bóthe, B., Rigó, A., Orosz, G., 2021. Toward an improved understanding of work motivation profiles. *Appl. Psychol.* 70 (3), 986–1017. <https://doi.org/10.1111/apps.12256>.
- Trang, S.T.N., Brendel, A., 2019. A meta-analysis of deterrence theory in information security policy compliance research. *Inf. Syst. Front.* 21 (6), 1265–1284. <https://doi.org/10.1007/s10796-019-09956-4>.
- Tursunbayeva, A., Pagliari, C., Di Lauro, S., Antonelli, G., 2022. The ethics of people analytics: risks, opportunities and recommendations. *Personnel Rev.* 51 (3), 900–921. <https://doi.org/10.1108/PR-12-2019-0680>. S.
- Van den Broeck, A., Howard, J.L., Van Vaerenbergh, Y., Leroy, H., Gagné, M., 2021. Beyond intrinsic and extrinsic motivation: a meta-analysis on self-determination theory's multidimensional conceptualization of work motivation. *Organ. Psychol. Rev.* 11 (3), 240–273.
- van Haastrecht, M., Yigit Ozkan, B., Brinkhuis, M., Spruit, M., 2021. Respite for SMEs: a systematic review of socio-technical cybersecurity metrics. *Appl. Sci.* 11 (15), 6909. <https://doi.org/10.3390/app11156909>.
- Vance, A., Siponen, M., Pahlila, S., 2012. Motivating IS security compliance: insights from habit and protection motivation theory. *Inf. Manag.* 49, 190–198. <https://doi.org/10.1016/j.im.2012.04.002>.
- Waqdan, M., Louafi, H., Mouhoub, M., 2025. Security risk assessment in IoT environments: a taxonomy and survey. *Comput. Secur.* 154, 104456. <https://doi.org/10.1016/j.cose.2025.104456>.
- Warrington, C., Syed, J., Tappin, R.M., 2021. Personality and employees' information security behavior among generational cohorts. *Comput. Inf. Sci.* 14 (1), 26–38. <https://doi.org/10.5539/cis.v14n1p26>.
- Warkentin, M., Willison, R., 2009. Behavioral and policy issues in information systems security: the insider threat. *Eur. J. Inf. Syst.* 18 (2), 101–105. <https://doi.org/10.1057/ejis.2009.12>.
- Weller, B.E., Bowen, N.K., Faubert, S.J., 2020. Latent class analysis: a guide to best practice. *J. Black Psychol.* 46 (4), 287–311. <https://doi.org/10.1177/0095798420930932>.
- Whitman, M.E., Mattord, H.J., 2018. *Principles of Information Security*, 6th ed. Cengage Learning.
- Workman, M., Bommer, W., Straub, D., 2009. The amplification effects of procedural justice on a threat control model of information systems security behaviours. *Behav. Inf. Technol.* 28 (6), 563–575. <https://doi.org/10.1080/01449290802556021>.
- Xue, B., Xu, F., Luo, X., Warkentin, M., 2021. Ethical leadership and employee information security policy (ISP) violation: exploring dual mediation paths. *Organ. Cybersecur. J. Pract. Process People* 1 (1), 5–23. <https://doi.org/10.1108/O CJ-02-2021-0002>.